DO NOW — RECALL

Lesson 050 · Queue<T>: FIFO and Service Order

Name: _____ Class: _____ Date: _____

1. What is stack Pop order?

2. What is Peek?

3. What does Enqueue do?

4. An empty queue executes enqueue(A), enqueue(B), enqueue(C), x=dequeue(), enqueue(x), then dequeue(). Trace the final queue from front to rear.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Last-in-first-out.

2. Model answer

Read an end item without removal.

3. Model answer

Adds to rear of queue.

4. Model answer

C, A. The first dequeue stores A in x; A joins the rear, then the final dequeue removes B.

By the end, you can…

• A queue is first-in, first-out (FIFO). Enqueue joins the rear; Dequeue removes from the front.

• Queues are suitable for ordered service requests such as print jobs or breadth-first exploration, if the queueing policy is FIFO.

• Peek inspects the front without removing. An empty Queue<T> throws on Dequeue/Peek unless the code guards against it.

Vocabulary — know these terms

FIFO — First In First Out.

Enqueue — Add at queue rear.

Dequeue — Remove from queue front.

front — Next item to be served.

NEW LEARNING

Read, annotate and explain

Observe

A queue is first-in, first-out (FIFO). Enqueue joins the rear; Dequeue removes from the front.

Explain

Queues are suitable for ordered service requests such as print jobs or breadth-first exploration, if the queueing policy is FIFO.

Apply

Peek does not remove. Guard Dequeue/Peek with Count>0. Queue<T> handles storage internally; an explicit circular array queue instead tracks front, rear and count and wraps indexes modulo capacity (revisited in Lesson 197).

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace a ticket queue with Jo then Kai; dequeue one.

Trace and justify

Jo is served first; Kai remains at the front.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Queue A,B,C; Dequeue once, then Peek. State removed and front value. [6]

Question 2

Write C# code to enqueue 2,4,6 into Queue<int>, then safely dequeue one. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare Stack<T> and Queue<T> for DFS versus BFS on a graph, explaining removal order. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A queue's Dequeue removes the newest item first.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.